

PostgreSQL Docker Backup & Restore Guide

VPS → Windows Local Docker Environment

This guide explains how to:

1. Take a PostgreSQL backup from a VPS Docker container.
2. Download it to a Windows machine.
3. Restore it into a local Docker PostgreSQL container.

Project Information

VPS Container

govtjobsnow_db

Local Docker Containers

govtjobsnow_app
govtjobsnow_db

Database

govtjobsnow

PostgreSQL User

postgres

PART 1: Create Backup on VPS

Create PostgreSQL Custom Backup (.backup)

SSH into VPS:

```
docker exec govtjobsnow_db pg_dump -U postgres -Fc govtjobsnow -f /tmp/  
govtjobsnow_backup.backup
```

Verify backup:

```
docker exec govtjobsnow_db ls -lh /tmp/govtjobsnow_backup.backup
```

Example output:

```
-rw-r--r-- 1 root root 341.1K Jun 9 19:36 /tmp/govtjobsnow_backup.backup
```

Copy Backup from Container to VPS Host

Sometimes `docker cp` may fail.

Reliable method:

```
docker exec govtjobsnow_db cat /tmp/govtjobsnow_backup.backup > /root/  
govtjobsnow_backup.backup
```

Verify:

```
ls -lh /root/govtjobsnow_backup.backup
```

PART 2: Download Backup to Windows

Run on Windows PowerShell:

```
scp root@72.60.101.23:/root/govtjobsnow_backup.backup D:\Backup
```

Example:

```
scp root@72.60.101.23:/root/govtjobsnow_backup.backup D:\Backup
```

Result:

```
D:\Backup\govtjobsnow_backup.backup
```

PART 3: Restore Backup into Local Docker

Verify Local Containers

```
docker ps
```

Example:

```
govtjobsnow_app  
govtjobsnow_db
```

Copy Backup into Local PostgreSQL Container

```
docker cp D:\Backup\govtjobsnow_backup.backup govtjobsnow_db:/  
govtjobsnow_backup.backup
```

Verify:

```
docker exec govtjobsnow_db ls -lh /govtjobsnow_backup.backup
```

Expected:

```
-rwxr-xr-x 1 root root 341.1K ...
```

Stop Application Container

Avoid active DB connections.

```
docker stop govtjobsnow_app
```

Terminate Existing Connections

Connect to `postgres` database and terminate connections to `govtjobsnow`.

```
docker exec govtjobsnow_db psql -U postgres -d postgres -c "SELECT
pg_terminate_backend(pid) FROM pg_stat_activity WHERE datname='govtjobsnow'
AND pid <> pg_backend_pid();"
```

Drop Existing Database

IMPORTANT: This deletes current local data.

```
docker exec govtjobsnow_db psql -U postgres -d postgres -c "DROP DATABASE IF
EXISTS govtjobsnow;"
```

Create Fresh Database

```
docker exec govtjobsnow_db psql -U postgres -d postgres -c "CREATE DATABASE
govtjobsnow;"
```

Restore Backup

```
docker exec govtjobsnow_db pg_restore -U postgres -d govtjobsnow /
govtjobsnow_backup.backup
```

PART 4: Verify Restore

List Tables

```
docker exec govtjobsnow_db psql -U postgres -d govtjobsnow -c "\dt"
```

Check Database Size

```
docker exec govtjobsnow_db psql -U postgres -d govtjobsnow -c "SELECT
pg_size_pretty(pg_database_size('govtjobsnow'));"
```

PART 5: Start Application

```
docker start govtjobsnow_app
```

Optional: Backup Local Database Before Restore

Always recommended.

```
docker exec govtjobsnow_db pg_dump -U postgres -Fc govtjobsnow >  
local_before_restore.backup
```

Troubleshooting

Error

```
cannot drop the currently open database
```

Cause:

You are connected to the database you are trying to drop.

Fix:

Connect to `postgres` database:

```
docker exec govtjobsnow_db psql -U postgres -d postgres
```

Then drop:

```
DROP DATABASE govtjobsnow;
```

Error

```
FATAL: terminating connection due to administrator command
```

Cause:

Expected when killing active database sessions.

Usually safe to continue.

Verify Backup Contents

```
docker exec govtjobsnow_db pg_restore -1 /govtjobsnow_backup.backup
```

This lists all objects contained in the backup.

Quick Restore Checklist

- ☐ Copy .backup file from VPS
- ☐ Download to Windows
- ☐ docker cp backup into container
- ☐ Stop app container
- ☐ Terminate DB connections
- ☐ Drop database
- ☐ Create database
- ☐ pg_restore backup
- ☐ Verify tables
- ☐ Start app container